

ETHAN SIMON

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Professional Summary

Junior Mechanical Engineer student seeking a research experience or internship. Prior experience in CAD modeling, simulation and data acquisition. Designed and implemented a flexural fatigue test machine as well as other stress and functionality testing. Design and built multicopter and Tail Sitter drones and integrated with electronic systems as well manufactured parts and created assembly jigs. Additionally, created simulations in MATLAB and Python to inform designs. Work experience developing a GUI for control of a liquid propulsion rocket test stand.

EDUCATION

Case Western Reserve University – 3.8 GPA

Cleveland, OH

Bachelor of Science in Mechanical Engineering

Aug 2024 – May 2028

- **Michalson Morley Scholar** – Merit Scholarship awarded to top performing students named after the famous CWRU experiment.
- **Case Alumni Association Scholarship**- recipient of scholarship in my sophomore year, given out to notable engineering students.
- **Relevant Coursework:** Dynamics, Thermodynamics, ODE's, PDE's, Design and Manufacturing I/II, Statics

PROFESSIONAL EXPERIENCE

Tylok International

Cleveland, OH

Mechanical Engineering Co-Op

Jan 2026 – Aug 2026

- Full design, manufacturing, and implementation of a flexural fatigue test machine for internal testing in accordance with ASTM standards to test a variety of parts and sizes, able to subject >2,500lbs of load, run up to 80,000 cycles, record all cycle data with strain gauges.
- Developed other internal stress and verification testing apparatus including thermal testing, flow testing, tensile testing, and pressure testing, using a variety of sensors, custom software, and custom machined set-ups.
- Worked with the sales team to improve product showcasing and system demonstrations for trade shows, which are in use now.
- Researched, acquired and integrated data acquisition with a Dewesoft system, developed and used programs in DewesoftX as well as C++ for technical analysis.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

VTOL CWRU

Aug 2024 – Present

Mechanical and Aerospace Team Lead, VFS DBVF 2025 champion

- Developed 10+ VTOL platforms including Hexacopters, Quadcopters, Tiltrotors and Tail-sitters.
- Used SolidWorks to develop parts and assemblies, as well as for analysis including vibrational analysis to identify harmonic resonance, as well as stress analysis to ensure FOS in the arm assemblies.
- Wrote a mission model script in MATLAB to find the best possible configuration for the DBVF competition.
- Adapted the MATLAB mission model into python and advanced the physics engine to use quaternions in transition areas, as well as updated the drag modeling for wings to include parasitic drag and stalling.
- Developed other simple projects such as a NACA airfoil tool that generates the SolidWorks curve geometry of any NACA airfoil.
- Developed project planning documents to instruct and guide new members, 7 key safety documents, and work instructions on how to use various machines, and software.

Liquid Propulsion Team

Aug 25 – Jan 26

- Developed a control GUI in PyQt5 that integrate the mechanical control with the software to run hot fire sequences
- Designed mechanical control systems to regulate thrust using solenoids and servos

University of Michigan Battery Lab

2022-2024

- Physical analysis and data collection of single and multi-cell batteries organized in spread sheets.
- Python based analysis of cells using libraries including Pandas, Scikit-learn, SciPy, and hdbscan

Sigma Chi Fraternity

- KTLW leadership conference – learned about leadership and how to help others to grow into leaders.
- Scholarship Chair: organized mentorship programs, connected members with resources, facilitated career workshops
- House Manager: coordinated housing of the chapter, helped develop housing corps, communicated with CWRU and other orgs.

SKILLS & INTERESTS

- **Skills:** SolidWorks, MATLAB, Python, Arduino, Microsoft Office Suite, QGroundControl, Machining, Mechanical Analysis
- **Interests:** Soccer, Fishing, Hiking, Cooking, Woodworking, Catan, Weight-Lifting, Music.